Block 4 – Session 2

Cloud Computing & Cloud Networking

Arquitectura de Xarxes

Universitat Pompeu Fabra

Adrián Pino <adrian.pino@upf.edu>

Outline

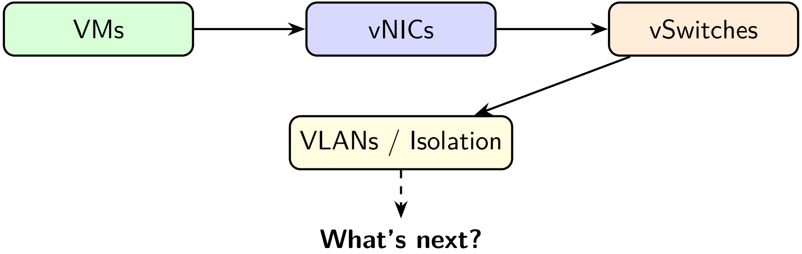
If We Can Virtualize Everything... Why Not Sell It?

Virtualization is the technology.
Cloud computing is the business model.

Learning objectives:

1. Understand cloud computing's definition and essential characteristics
2. Distinguish IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service) service models
3. Design a basic cloud virtual network with subnets and security controls

Recap – What We Virtualized (Session 1)



- Compute virtualization: VMs and containers
- Network virtualization: vNICs, vSwitches, vRouters, VXLAN overlays
- Together: the **building blocks** of the cloud

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	**Cloud Computing**
Run multiple VMs on one host	Offer compute as a **service**
Internal IT optimization	External/internal consumption
Manual provisioning possible	**Automated, self-service**
You own the hardware	Provider owns the hardware

Key difference: **automation and self-service** (no phone calls, no tickets, no waiting).

Discussion: From Virtualization to Cloud

Your company already runs VMs on its own servers. Why would you move to the cloud instead of keeping everything in-house?

Hints: think about CapEx vs OpEx, scaling speed, and who manages what.

Outline

Cloud Computing – Definition (NIST: National Institute of Standards and Technology)

> *A model for enabling ubiquitous, convenient, **on-demand network access** to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.*

Source: NIST SP 800-145 (2011). Still the standard definition used industry-wide.

Five Essential Characteristics

Characteristic	What it means
On-demand self-service	Provision resources without human interaction
Broad network access	Available over the network from any device
Resource pooling	Shared infrastructure, multi-tenant
Rapid elasticity	Scale up/down automatically
Measured service	Pay only for what you use (metered)

- **All five** must be present to qualify as "cloud computing"
- If you need to call someone to get a VM $\rightarrow$ it is not cloud

Source: NIST SP 800-145, Mell & Grance, 2011.

On-Demand Self-Service

- Users provision resources through a **portal or API (Application Programming Interface)**
- No phone calls, no tickets, no waiting for approval
- Example: launch a VM in **30 seconds** via a web console

Analogy: vending machine (cloud) vs restaurant with a waiter (traditional IT).

Broad Network Access + Resource Pooling

Broad network access:

- Resources accessible from **any device** with network connectivity
- Standard protocols: HTTPS, SSH, REST APIs
- No proprietary client software needed

Resource pooling:

- Provider's resources serve **multiple tenants** simultaneously
- Tenants do not know (or care) where their resources are physically located
- Dynamically assigned based on demand
- Builds on the **multi-tenancy** concepts from Session 1

Rapid Elasticity + Measured Service

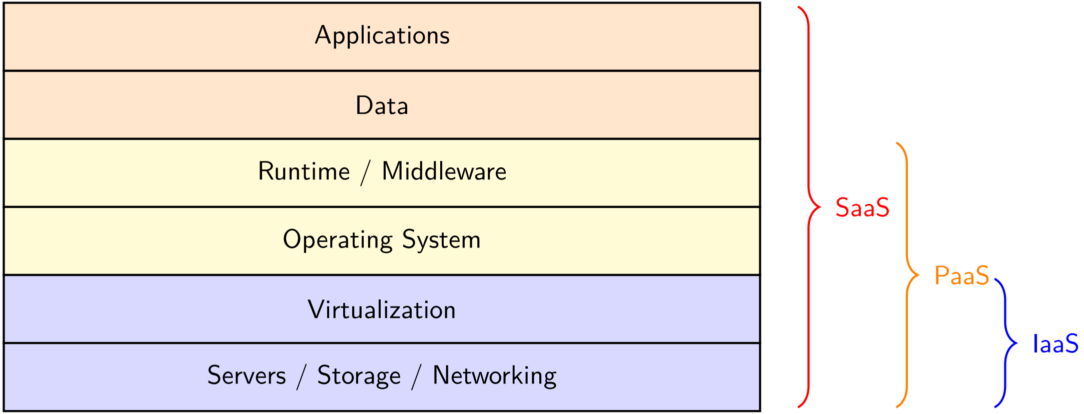
Rapid elasticity:

- Resources appear **unlimited** to the consumer
- Scale out (add) and scale in (remove) automatically
- Example: web store on Black Friday $\rightarrow$ $10\times$ servers, then back to normal

Measured service:

- Usage is **metered** (CPU-hours, GB stored, GB transferred)
- **Pay-per-use** model $\rightarrow$ OpEx instead of CapEx
- Transparent monitoring and billing reports

Cloud Service Models – The Stack



IaaS – Infrastructure as a Service

- Provider gives you: **VMs, storage, networks**
- You manage: OS, middleware, applications, data
- **Maximum control, maximum responsibility**

Examples:

- **AWS EC2 (Elastic Compute Cloud)**: virtual machines
- **Azure Virtual Machines**
- **Google Compute Engine**

Use case: "I want a Linux server in Frankfurt, 4 CPUs, 16 GB RAM, right now."

Source: AWS EC2 Documentation, docs.aws.amazon.com.

PaaS – Platform as a Service

- Provider gives you: **runtime environment + tools**
- You manage: only your **application code and data**
- No need to manage OS, patches, or scaling infrastructure

Examples:

- **Google App Engine**
- **AWS Elastic Beanstalk**
- **Heroku**

Use case: "I wrote a Python web app. Just run it, scale it, I do not care about servers."

SaaS – Software as a Service

- Provider gives you: a **complete application**
- You manage: your **data and user configuration**
- No installation, no maintenance, no updates to worry about

Examples:

- **Google Workspace** (Gmail, Docs, Drive)
- **Microsoft 365** (Word, Excel, Teams)
- **Salesforce**, **Slack**, **Zoom**

Use case: "I need email for 500 employees. I do not want to run a mail server."

Service Models – Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider
Hardware	Provider	Provider	Provider

- IaaS $\rightarrow$ SaaS: **less control, less responsibility**
- Major providers (AWS, Azure, GCP) offer all three models

Discussion: Cloud Fundamentals

A university wants to offer a web-based tool
for students to submit assignments.
Should they choose IaaS, PaaS, or SaaS? Why?

Hints: consider the IT team size, maintenance burden, and how much control they need.

Outline

Putting It All Together

- In Session 1 we saw the **building blocks**: vSwitches, vRouters, vNICs, VLANs
- These components do not run in isolation; they are managed by **platforms**
- Platforms orchestrate **VMs** or **containers** and configure virtual networking under the hood

Two categories:

- **Cloud platforms**: manage infrastructure at scale
  - Compute, storage, networking, and **VMs**
- **Container orchestration platforms**: deploy and scale **containerized** applications

Cloud Platforms

- **Self-managed / Private Cloud**: you own or rent the servers, install and manage the virtualization software yourself. Full control, but requires dedicated staff and expertise.

Platform	**Description**	**Logo**
OpenStack	Open-source, widely used in private clouds	
VMware vSphere	Enterprise leader, proprietary	
Proxmox VE	Open-source, lightweight alternative	

- **Managed / Public Cloud**: rent capacity on demand, pay-as-you-go. Provider handles hardware, networking, cooling, security, and updates.

Provider	**Description**	**Logo**
Amazon Web Services	Market leader since 2006	
Microsoft Azure	Strong enterprise integration	
Google Cloud Platform	Data and AI focus	

These map directly to the IaaS model we just covered: the provider (or you) manages compute, storage, and networking.

Container Orchestration Platforms

- **Self-hosted / Private**: you install and operate on your own servers

Platform	Description	Logo
Docker	Container runtime; works well on a single host Managing hundreds of containers across hosts? Unmanageable	docker
Kubernetes (K8s)	Created to solve this: scheduling, scaling, networking, self-healing	kubernetes
OpenShift	Enterprise K8s by Red Hat; adds security, CI/CD, UI	Red Hat OpenShift

- **Hosted / Public**: cloud providers offer the same platforms as managed services. Each provider uses different names for essentially the same product (e.g., AWS ECS/EKS, Azure AKS, Google GKE, OpenShift on all major clouds).

Discussion: Platforms

A startup with 5 engineers needs to deploy a web application that may go from 100 to 100,000 users. Would you build a private cloud or use a public one? Why?

Hints: team size, upfront cost, time to market, scaling needs, control over infrastructure.

Outline

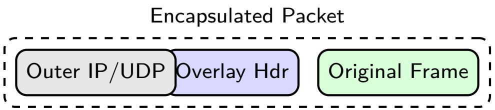
The VLAN Scalability Problem

- In Session 1 we used **VLANs** to isolate tenants on virtual networks
- VLANs use a **12-bit ID** → maximum **4,096** virtual networks
- Cloud providers host **millions** of tenants → VLANs are not enough
- Additional limitations:
  - VLANs are confined to a **single L2 domain**
  - Moving a VM to another host may require VLAN reconfiguration

We need a technology that scales beyond 4K networks and works across L3 boundaries.

Overlay Networks: Concept

- An **overlay network** is a virtual network built **on top of** the physical one
- Uses **encapsulation**: wrap virtual frames inside physical packets



- The physical network only sees the **outer headers**
- The virtual (tenant) traffic is hidden inside → **full isolation**

VXLAN: Overview

- **VXLAN** (Virtual eXtensible LAN): most widely used overlay protocol
- Encapsulates L2 frames in **UDP packets** over the physical network
- Uses a **24-bit VNI** (VXLAN Network Identifier)
  - 2^{24} = **16 million** virtual networks (vs 4,096 VLANs)
- Endpoints: **VTEPs** (VXLAN Tunnel Endpoints)
  - Encapsulate at source, decapsulate at destination

Source: IETF RFC 7348, "Virtual eXtensible Local Area Network (VXLAN)," 2014.

VXLAN: How It Works

- VM A sends a frame → VTEP 1 wraps it in UDP with a VNI
- Travels over the physical network as a regular UDP packet
- VTEP 2 strips outer headers, delivers original frame to VM B
- VMs on the **same VNI** form an isolated L2 domain

Overlay Benefits for Scalability

- **16 million virtual networks** (vs 4,096 VLANs)
- Physical network does not need to know about tenants
- VMs can **migrate** across hosts without reconfiguring the underlay
- Decouples **virtual topology** from physical topology

This is the foundation of cloud networking. Next: how providers build **virtual networks** on top of overlays.

Source: IETF RFC 8926, “Geneve: Generic Network Virtualization Encapsulation,” 2020.

Discussion: Overlay Networks

Why can the physical network remain simple
if we use overlay networks?
What are the trade-offs of encapsulation?

Hints: think about overhead (extra headers), MTU implications, and troubleshooting complexity.

Outline

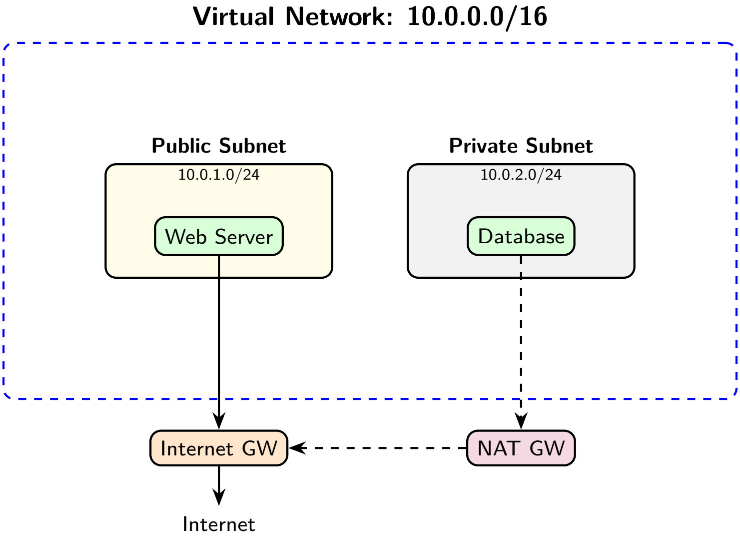
Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
  - AWS: Virtual Private Cloud (VPC); Azure: Virtual Network (VNet); GCP: VPC Network
- Think of it as your **private data center network** in the cloud
- You define:
  - **Address space** (e.g., `10.0.0.0/16`)
  - **Subnets** (subdivisions of the address space)
  - **Routing rules** and **security policies**

Key: virtual networks are built on top of the **overlay network** concepts we just covered.

Source: AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Network Architecture



Subnets – Public vs Private

Public subnet:

- Has a route to an **Internet Gateway**
- Instances can have **public IP addresses**
- Accessible from the Internet (if security rules allow)
- Use for: web servers, load balancers, bastion hosts

Private subnet:

- **No direct route** to the Internet
- Instances only have **private IPs**
- Can access Internet outbound via **NAT Gateway**
- Use for: databases, internal services, application backends

Route Tables

- Each subnet is associated with a **route table**
- Route table = set of rules determining where traffic goes

Destination	Target	Subnet type
`10.0.0.0/16`	local (within virtual network)	Both
`0.0.0.0/0`	Internet Gateway	Public
`0.0.0.0/0`	NAT Gateway	Private

- **Local route**: traffic within the virtual network stays inside (always present)
- **Default route** (`0.0.0.0/0`): where to send everything else
- Public subnets → IGW; Private subnets → NAT GW

Internet Gateway vs NAT Gateway

Internet Gateway (the front door):

- **Bidirectional**: the world can reach your instance, and it can reach the world
- Instance needs a **public IP**
- Use for: web servers, load balancers, anything public-facing

NAT Gateway (the fire exit):

- **One-way**: your instance can reach the Internet (e.g., download updates, call APIs), but nobody outside can reach it directly
- Instance keeps a **private IP** only
- Use for: databases, backends that need to download patches but must stay hidden

Same **NAT concept from Block 3**, now as a managed cloud service.

Security Groups – Instance-Level Firewall

- **Security group** = virtual firewall attached to an instance
- Rules specify allowed **inbound** and **outbound** traffic
- **Stateful**: allow inbound → response automatically allowed

Direction	**Protocol**	**Port**	**Source/Dest**
Inbound	TCP	80	`0.0.0.0/0`
Inbound	TCP	443	`0.0.0.0/0`
Inbound	TCP	22	`10.0.0.0/16`
Outbound	All	All	`0.0.0.0/0`

Source: AWS, "Security Groups for Your VPC," docs.aws.amazon.com.

Network ACLs (Access Control Lists) – Subnet-Level Firewall

- **Network ACL (Access Control List)** = firewall at the **subnet** level
- Rules evaluated in **order** (numbered, first match wins)
- **Stateless**: must explicitly allow both inbound AND outbound

Rule #	Direction	Protocol	Port	Action
100	Inbound	TCP	80	ALLOW
200	Inbound	TCP	443	ALLOW
*	Inbound	All	All	DENY

Source: AWS, "Network ACLs," docs.aws.amazon.com.

Security Groups vs Network ACLs

	Security Groups	Network ACLs
Scope	Instance level	Subnet level
State	Stateful	Stateless
Rules	Allow only	Allow + Deny
Evaluation	All rules evaluated	First match wins
Default	Deny all inbound	Allow all

Defense in depth: use both together.

- Security Groups → fine-grained per-instance control
- Network ACLs → broad subnet-level guardrails

Discussion: Cloud Networking

You deploy a web application in your cloud virtual network. The web server needs to be public, but the database must not be reachable from the Internet. How?

Hints: think about public vs private subnets, security groups, and NAT gateways.

Outline

Key Takeaways

1. Virtualization is the **technology**; cloud is the **business model**
2. Five NIST characteristics define true cloud computing
3. **IaaS / PaaS / SaaS** differ in who manages what
4. **VXLAN** overlays scale isolation to 16M networks (vs 4K VLANs)
5. A **virtual network** is your isolated cloud network, built on top of overlays
6. **Security groups** (instance) + **ACLs** (subnet) = defense in depth

Discussion

You are designing a cloud network for a company with public web servers and private databases. Which components would you use and why?

Think about: public vs private subnets, security groups, NAT gateways, route tables.

References

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